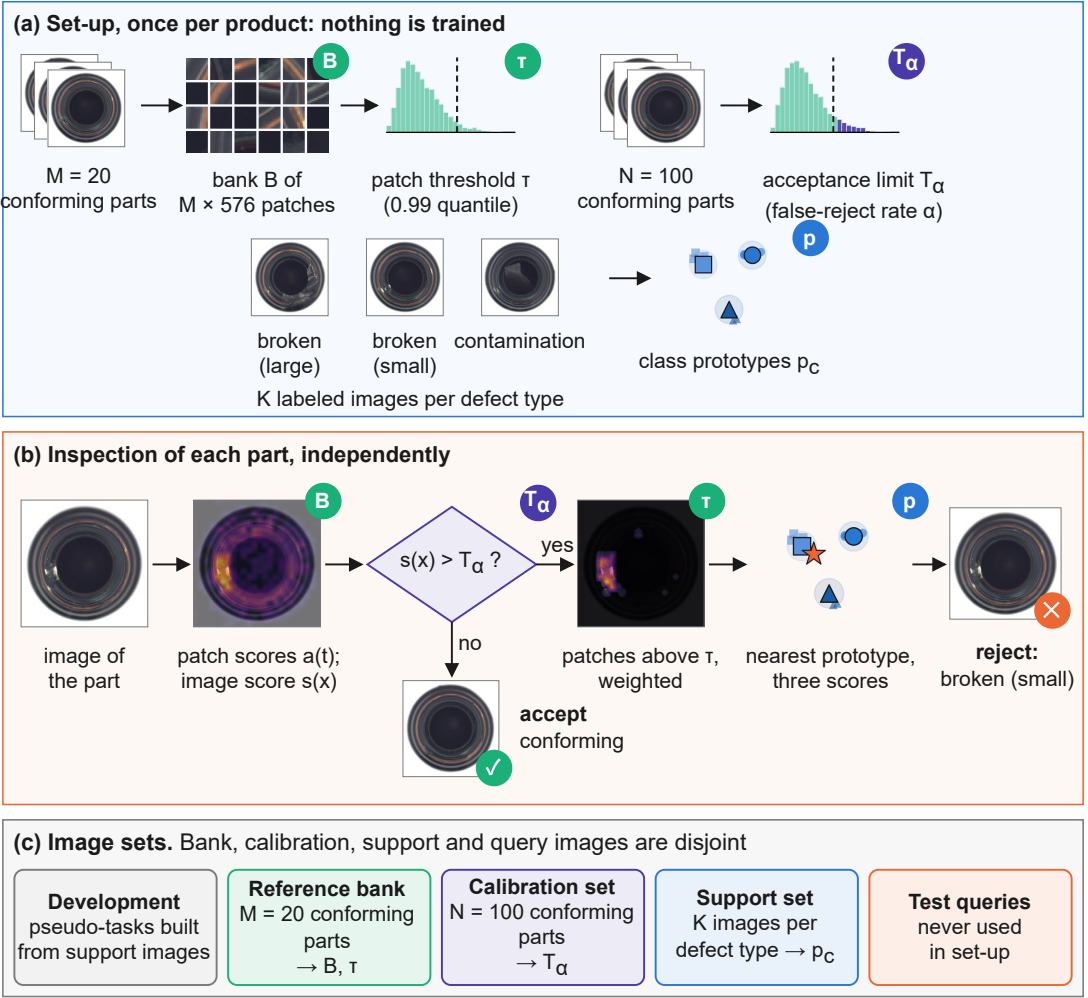

(a) Set-up, once per product: nothing is trained
B
τ
T_α
M = 20
conforming parts
bank B of
M × 576 patches
patch threshold τ
(0.99 quantile)
N = 100
conforming parts
acceptance limit T_α
(false-reject rate α)
p
broken
(large)
broken
(small)
contamination
class prototypes p_c
K labeled images per defect type
(b) Inspection of each part, independently
B
T_α
τ
p
s(x) > T_α ?
yes
no
image of
the part
patch scores a(t);
image score s(x)
patches above τ,
weighted
nearest prototype,
three scores
reject:
broken (small)
accept
conforming
(c) Image sets. Bank, calibration, support and query images are disjoint
Development
pseudo-tasks built
from support images
Reference bank
M = 20 conforming
parts
→ B, τ
Calibration set
N = 100 conforming
parts
→ T_α
Support set
K images per
defect type → p_c
Test queries
never used
in set-up